

Long-read best practices

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Long-read technology

- ▶ Sequencing method that produces reads > 10kb, sometimes even megabases.
- ▶ PacBio and Oxford Nanopore have developed technologies for accurate long-read capturing.
- ▶ Used for detecting structural variants, full-length transcripts/isoforms, genome assembly, etc.
- ▶ Long-read is generally less accurate and more expensive than short read, but technology is improving

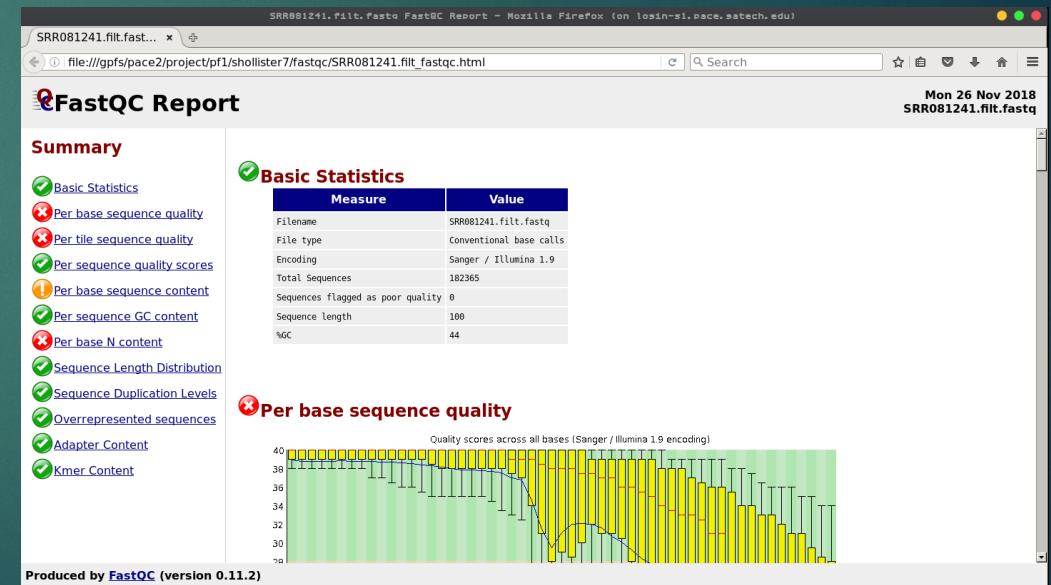
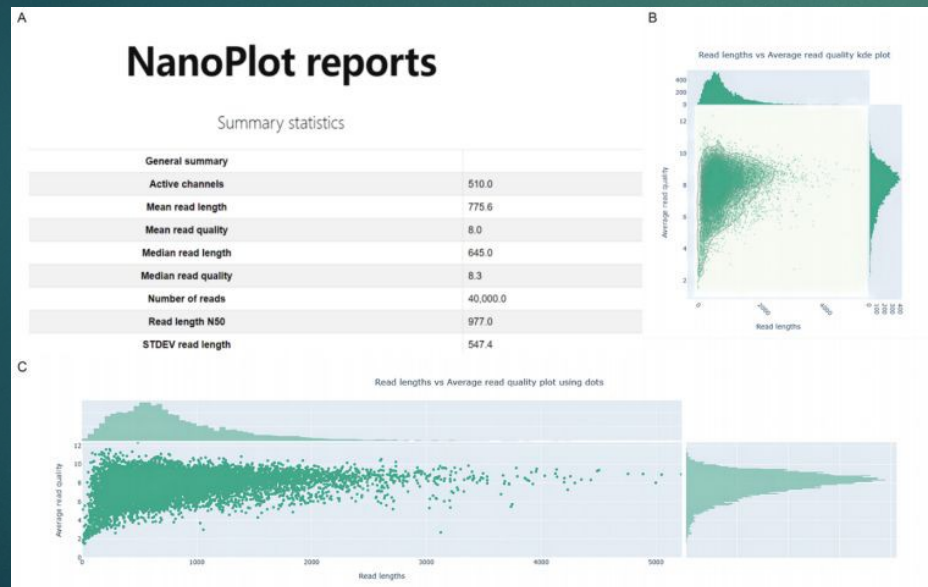
Oxford Nanopore vs PacBio

- ▶ Oxford Nanopore is used when you need very long reads
 - ▶ For example, structural variants are better studied using long reads
- ▶ PacBio is used when you need very high accuracy
 - ▶ SNP/INDEL calling, genome assembly

Feature	Oxford Nanopore	PacBio
Read Length	Very long (Megabases)	Long (>10kb)
Accuracy	90-98% accuracy	>99% accuracy
Sequencing Method	Real-time	Consensus (Read is read multiple times)
Cost per base	Lower, but generally more bases	Higher due to accuracy

Quality control

- ▶ Nanoplot – QC tool specifically made for long-read data.
- ▶ FastQC – can still be used for long-read data, but typically used for short-read data.



Alignment for long-read RNA

- ▶ Minimap2 is dominant tool for long read RNA alignment.
- ▶ Benchmark studies mainly look at transcript level tools or assembly tools
 - ▶ Transcript assemblers, quantifiers, differential tools
- ▶ [LoTempio et al](#) – Benchmarks the alignment tools for long-read RNA
 - ▶ Minimap2 outperforms all of the other tools
 - ▶ Winnowmap2 was the fastest tool
- ▶ Alignment tool generally less important in long-read because the longer reads are often easier to align to the genome.

Dong et al. Benchmark study for long-read RNA analysis

- ▶ Isoform Reconstruction and Detection
 - ▶ StringTie2: Highest precision (predictions were correct)
 - ▶ Bambu: Highest power (more novel isoforms were detected)
 - ▶ FLAIR, TALON, cupcake all overpredicted – a lot of artifactual isoforms
- ▶ Differential transcript expression (DTE)
 - ▶ DESeq2, edgeR, limma all effective and performed similarly in differential transcript analysis
- ▶ Differential transcript usage (DTU)
 - ▶ DRIMSeq and DEXSeq were the most powerful DTU tools but had high false discovery rates; edgeR controlled FDR well but had the lowest power

Pardo-Palacios et al – Benchmark study for long-read transcript identification

- ▶ All reads were aligned using minimap2
- ▶ Read quality and length, not quantity were the drivers of accurate transcript reconstruction
- ▶ Tools for transcript identification varied in performance
 - **Best Tools for reference based: Bambu and IsoQuant, FLAMES.**
 - **De Novo Transcript Detection – StringTie2** for assembly and **Isoquant** for quantification, **RNA-Bloom** and **rnaSPAdes** both prone to overcalling.

Dierckxsens et al – Benchmark study for long-read structural variant

- ▶ **cuteSV, SVIM, and pbsv** performed best overall, with pbsv being most accurate for SV position and length.
- ▶ **combiSV**, a new tool, combines outputs from 6 SV callers to improve recall, precision, and accuracy.
- ▶ Higher sequencing **depth doesn't always help**, but **longer reads** consistently improve SV detection.

Luth et al. – Genome assembly tools with long-read data

- ▶ **Flye** is the top-performing long-read assembler overall for PacBio CLR and ONT reads.
- ▶ **Hifiasm** and **LJA** are best for high-accuracy PacBio HiFi assemblies.
- ▶ **Miniasm**, **Raven**, and **wtdbg2** work well for small genomes with lower resource use.
- ▶ Longer reads improve reduce interruptions in large genomes but don't always enhance identity or gene recovery.

Variant Callers for long-read DNA data

- ▶ Clair3
 - ▶ Deep learning-based
 - ▶ Top accuracy on ONT & PacBio HiFi
- ▶ PEPPER-Margin-DeepVariant
 - ▶ Ensemble model (PEPPER + DeepVariant)
 - ▶ Excellent for PacBio HiFi
- ▶ Mutserve2
 - ▶ Best for mitochondrial low-frequency variants
 - ▶ High sensitivity & F1 score

Differences from short-read

- ▶ Data comes in FAST5 format, not FASTQ.
 - ▶ Tools like Guppy and Bonito are used for base calling
- ▶ Trimming
 - ▶ Tools like FiltLong or NanoFilt are used to trim long-read data
- ▶ Difference in tools for variant calling and transcript identification/quantification
 - ▶ As discussed, different tools should be used when dealing with long-read data